Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · AI-Native Internal Tooling · Agentic Workflows · Python · FastAPI · Production Platform Engineering · CI/CD · LangGraph · AWS · Fintech

+1-703-261-5410 | lakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Senior Python Engineer who builds AI-native internal tooling and platform capabilities for engineering workflows - exactly the pattern Plaid's AI Tooling team needs. Built AgentProbe from scratch to solve a real internal pain point: engineers had no systematic way to measure AI output quality or catch regressions before production. The solution: a 15-endpoint FastAPI platform with a 5-metric eval engine, LLM-as-judge scoring, and a CI-integrated regression gate - zero vendor dependency, zero regressions shipped. Built LeaseI because a compliance investigation that should take 30 minutes was taking a full day: an 8-agent LangGraph platform with agentic workflows, auth-controlled HITL escalation, and deterministic guardrails for regulated outputs. Both systems went from ambiguous pain point → MVP → production-hardened platform with full ownership. At Entrata, owned and operated internal monitoring tools that reduced triage time 40%, built and maintained production platforms at 99.5% uptime for 100K+ users over 7 years. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Engineering & Platform: Python (8yr), JavaScript, TypeScript, FastAPI, Flask, SQL, Docker, Microservices, REST APIs, CI/CD (GitHub Actions), AWS (S3, EC2, RDS), RabbitMQ, Redis, PostgreSQL, MySQL, Oracle, Bash, Linux, Git

AI & Agentic Systems: LangGraph, LangChain, Agentic Workflows, Multi-Agent Orchestration, RAG Pipelines, LLM Orchestration, ChromaDB, Model Evaluation, LLM-as-Judge, Hallucination Detection, HITL, Auth-Controlled Escalation, Prompt Engineering, OpenAI API, Anthropic API, MCP

Observability & Quality: Internal Tooling Platforms, Production Monitoring, CI-Integrated Regression Gating, End-to-End QA Ownership, Agile/Scrum, scikit-learn, PyTorch, BERT, spaCy, NumPy, Pandas, Tableau

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Translated an ambiguous operational workflow (mission planning) into a production-hardened Python agentic system - rapid MVP to validated platform with end-to-end automated testing; designed auth-controlled validation guardrails with deterministic compliance checks before delivery.
- Built and operated RAG retrieval pipeline over distributed data sources; observability on retrieval quality and response groundedness across edge cases.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built Python and R ML pipelines on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end platform from ingestion through feature engineering, model evaluation, and validation across 190+ countries.
- Mentored 600+ students in Python, SQL, and software design; designed internal monitoring dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS
Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Built and operated internal tools that became production-critical** - internal debugging and monitoring platform reducing triage time 40%; owned the tooling end-to-end through incidents, upgrades, and multi-team adoption; maintained sub-10 bug count for 6+ consecutive months at 99.5% uptime, 100K+ monthly active users.
- **Designed platform constraints for production safety** - RabbitMQ async processing with retry, idempotency, and dead-letter guarantees; Redis caching; Docker microservices redesign (50% scalability improvement, 30% deployment reduction); AWS S3, EC2, RDS at production scale; 200+ SQL query optimizations (30% gain).
- **End-to-end ownership across 20+ scalable production modules** - Python/FastAPI backends, PostgreSQL/Oracle databases, REST APIs; 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents); Python/Bash automation eliminating 80% of manual operational workload.
- Translated ambiguous stakeholder requirements into durable software; mentored 5+ engineers; 300+ code reviews; cross-functional collaboration across product, design, and infrastructure teams.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL); data pipeline for 1.6M HUD/Tax Credit records with validation and reconciliation; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: AI-Native Internal Tooling Platform AI Tooling · Platform

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Built from ambiguous pain point to production platform: engineers had no internal tooling to detect AI output regressions before merge. Solved with a 15-endpoint FastAPI service, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge scoring, and CI-integrated regression gate - zero vendor dependency. 1,240+ test cases; zero regressions shipped to production.

Leasel: 8-Agent Agentic Internal Investigation Platform Agentic · Platform

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Translated a real internal workflow pain point (compliance investigations taking a full day) into a durable AI platform: 8-agent LangGraph stateful system, agentic workflows with auth-controlled HITL escalation, multi-source RAG with deterministic guardrails for regulated outputs. 93.6% precision, 87.9% F1 across 1,240 test cases; CI quality gate with full observability.

AI on FHIR: NLP-Driven Medical Records Query Platform Platform · Auth

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT | github.com/venkatalakshmiparimala/AI_on_FHIR

Internal tooling for non-engineer clinical users: NLP platform translating natural-language queries into FHIR API calls via BERT and LLMs. Reduced retrieval time ~60%; HIPAA-compliant RBAC with full auth boundaries and data isolation across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017